

# Fashion-MNIST Image Classification with PyTorch

## MLP, CNN and Hyperparameter Experiments — Week 2 Technical Report

### 1. Introduction

This project classifies clothing images from the Fashion-MNIST dataset using PyTorch. The work has three stages. First, a feed-forward neural network (MLP) was built as a baseline (Task 2.1). Second, a Convolutional Neural Network (CNN) was built and compared with the MLP (Task 2.2). Third, the CNN's hyperparameters were varied to see how test accuracy changed (Task 2.3). PyTorch was used because it provides ready-made tools for datasets, DataLoader, neural-network layers, loss functions and optimizers.

### 2. Dataset and Preprocessing

Fashion-MNIST contains grayscale images of clothing items. Each image is  $28 \times 28$  pixels with one channel, and there are 10 classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag and Ankle boot. The dataset has 60,000 training images and 10,000 test images. Figure 1 shows sample training images.

The data was loaded with `torchvision.datasets.FashionMNIST` and converted to tensors with `transforms.ToTensor()`. The data was then fed to the model in mini-batches by a PyTorch DataLoader, with shuffling for training and no shuffling for testing. The batch size was 64, giving 938 training batches and 157 test batches (from the Task 2.1 notebook output).

**Preprocessing difference between notebooks.** The MLP notebook (Task 2.1) used `ToTensor()` followed by `Normalize((0.5,), (0.5,))`. The CNN notebooks (Tasks 2.2 and 2.3) used `ToTensor()` only, with no normalization. This is a difference in the actual code, so the MLP–CNN comparison below is not controlled for preprocessing.

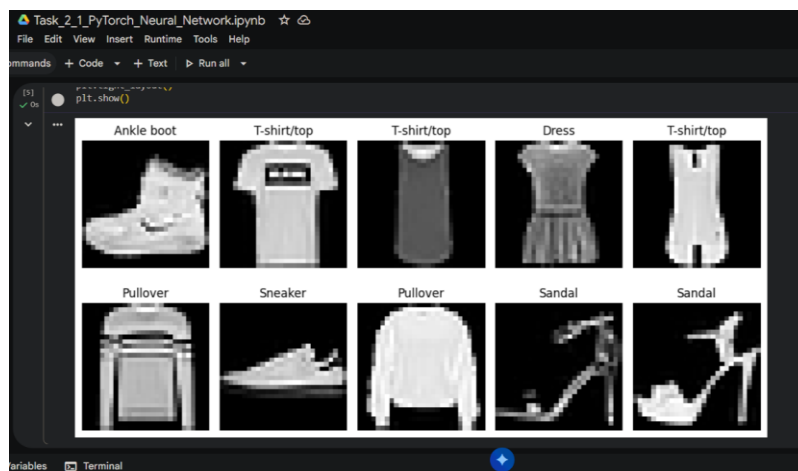


Figure 1. Sample Fashion-MNIST images used in the classification task.

### 3. Task 2.1 — MLP

Each  $28 \times 28$  image was flattened into a vector of 784 features, because a fully connected (Linear) layer takes a one-dimensional input. The architecture was:

Input (784)  $\rightarrow$  Linear 784 $\rightarrow$ 128  $\rightarrow$  ReLU  $\rightarrow$  Linear 128 $\rightarrow$ 64  $\rightarrow$  ReLU  $\rightarrow$  Linear 64 $\rightarrow$ 10

Training used CrossEntropyLoss, the Adam optimizer, learning rate = 0.001, batch size = 64 and 10 epochs. The training loss decreased from 0.5208 in epoch 1 to 0.2381 in epoch 10 (Figure 2). The trained model was saved as `fashion_mnist_mlp.pth`.

**Result: the MLP test accuracy was 88.31%.**

The classification report (Figure 3) shows the weakest per-class F1-score for Shirt (0.70). The strongest were Trouser (0.98) and Bag (0.97). The confusion matrix is shown in Figure 4.

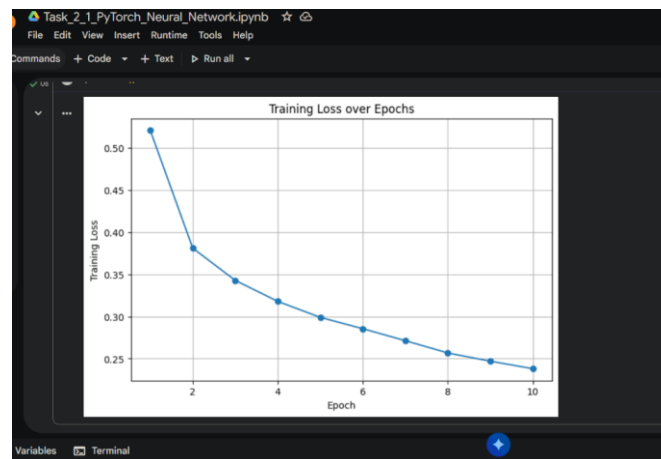


Figure 2. MLP training loss over 10 epochs.

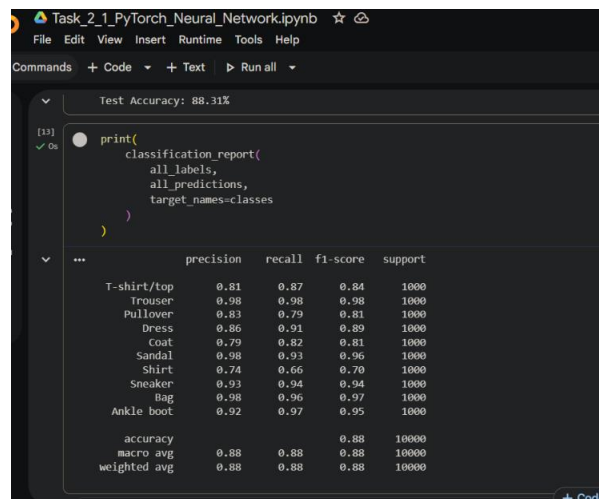


Figure 3. MLP classification report on the Fashion-MNIST test set.

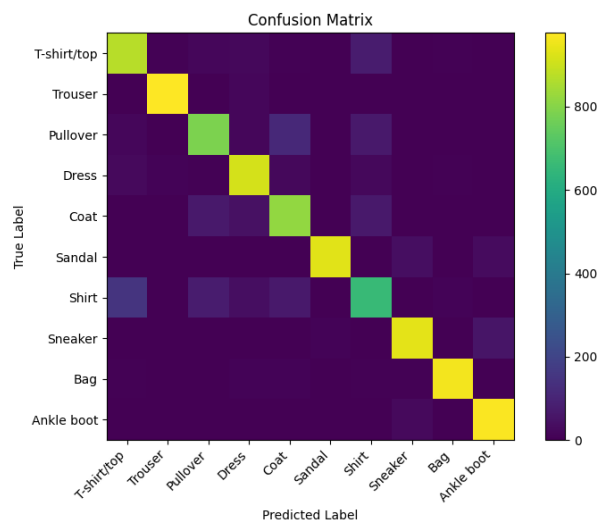


Figure 4. MLP confusion matrix on the Fashion-MNIST test set.

## 4. Task 2.2 — CNN

A CNN was used because convolution layers work on the 2D image directly, and pooling layers reduce the spatial size. The implemented architecture was:

Conv2D 1→32 (kernel 3×3, padding 1) → ReLU → MaxPool (2) → Conv2D 32→64 (kernel 3×3, padding 1) → ReLU → MaxPool (2) → Flatten → Linear 64×7×7→128 → ReLU → Linear 128→10

Training used CrossEntropyLoss, Adam, learning rate = 0.001, batch size = 64 and 10 epochs. The training loss decreased from 0.4612 to 0.1049 (Figure 5). The model was saved as fashion\_mnist\_cnn.pth.

**Result: the CNN baseline test accuracy was 91.92%.**

| Model        | Test Accuracy          |
|--------------|------------------------|
| MLP          | 88.31%                 |
| CNN baseline | 91.92%                 |
| Difference   | 3.61 percentage points |

In this experiment, the CNN achieved a higher test accuracy than the MLP. This is a result from this project only, and it does not show that CNNs are better in general. Shirt again had the lowest F1-score for the CNN (0.78, compared with 0.70 for the MLP) (Figure 6). The confusion matrix is shown in Figure 7.

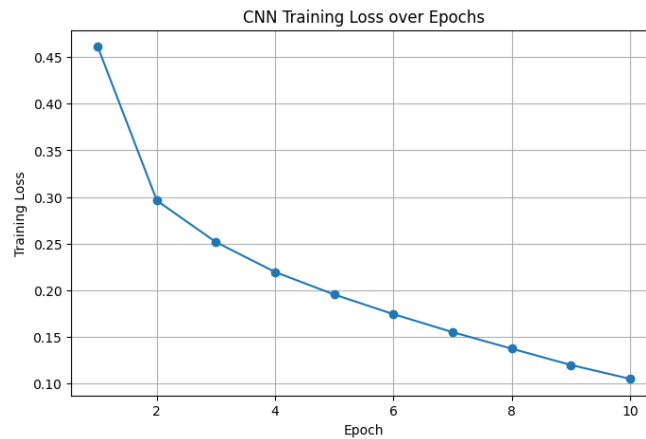


Figure 5. CNN training loss over 10 epochs.

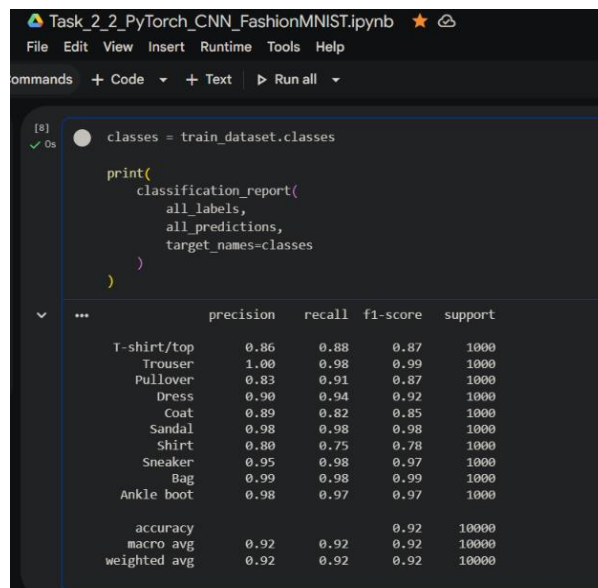


Figure 6. CNN classification report on the Fashion-MNIST test set.

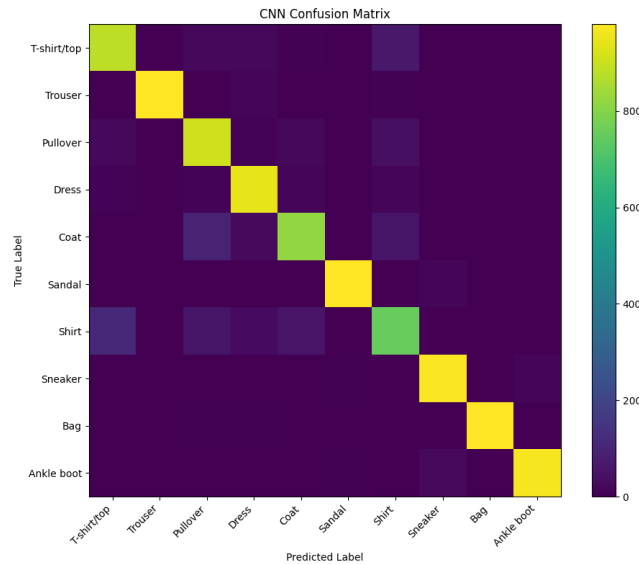


Figure 7. CNN confusion matrix on the Fashion-MNIST test set.

## 5. Task 2.3 — Hyperparameter Experiments

The purpose of Task 2.3 was to investigate how changing selected hyperparameters affected CNN test accuracy. Only one setting was changed at a time relative to the baseline.

- Experiment 1: a lower learning rate (0.0005).
- Experiment 2: a higher learning rate (0.002).
- Experiment 3: a larger batch size (128).
- Experiment 4: a larger number of epochs (20).

| Experiment   | Learning Rate | Batch Size | Epochs | Test Accuracy        | Difference from Baseline |
|--------------|---------------|------------|--------|----------------------|--------------------------|
| MLP          | 0.001         | 64         | 10     | 88.31%               | —                        |
| CNN Baseline | 0.001         | 64         | 10     | 91.92%               | —                        |
| Experiment 1 | 0.0005        | 64         | 10     | 91.69%               | −0.23 pp                 |
| Experiment 2 | 0.002         | 64         | 10     | 91.78%               | −0.14 pp                 |
| Experiment 3 | 0.001         | 128        | 10     | 91.36%               | −0.56 pp                 |
| Experiment 4 | 0.001         | 64         | 20     | [VALUE NOT PROVIDED] | [VALUE NOT PROVIDED]     |

Experiment 4 was conducted with 20 epochs, but its final test accuracy was not available in the provided evidence.

## 6. Results and Discussion

The MLP reached 88.31% and the CNN baseline reached 91.92%, a difference of 3.61 percentage points. Among the tested configurations with known results, the CNN baseline achieved the highest test accuracy (91.92%).

All three changed configurations scored slightly lower than the baseline. The largest drop was for batch size 128 (−0.56 pp). The learning-rate changes produced smaller drops (−0.23 pp for 0.0005 and −0.14 pp for 0.002). Each configuration was run once. No statistical significance test was performed, so these small

differences cannot be called meaningful. They also do not show that the baseline settings are optimal in general.

## **7. Conclusion**

An MLP and a CNN were implemented in PyTorch for Fashion-MNIST classification. The MLP reached 88.31% test accuracy and the CNN baseline reached 91.92%. Three hyperparameter experiments (two learning rates and one batch size) gave 91.69%, 91.78% and 91.36%, all slightly below the baseline. The Experiment 4 result is not available in the provided evidence, so no conclusion about training for 20 epochs is drawn. The project shows why changing one hyperparameter at a time and recording the results is a useful way to compare configurations.